



Solar Hackathon Challenge

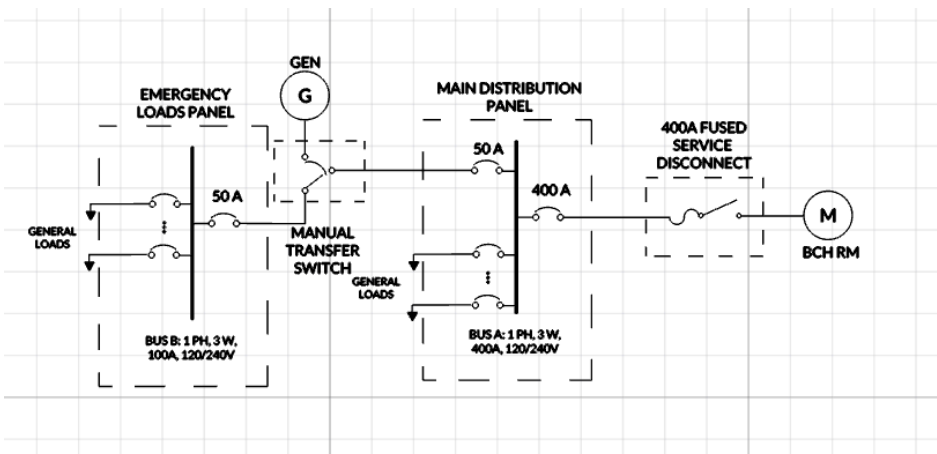
Challenge

Design a solar and battery system using **EG4 FlexBOSS21 inverters**, **EG4 280Ah All-Weather batteries**, and **GridBOSS units** to power a **400A residential service** during grid outages.

You are restricted to using the industry-standard **JA Solar JAM54D41-440/LB 440W solar panel** and must incorporate the home's existing **6kW generator (BE7500ID)** into the design.

For the purposes of this challenge, assume the homeowner's goal is to operate **completely off-grid**, with **no power exported or sold back to the utility grid**.

Below is a single-line diagram of the existing electrical system.





System Design Requirements

Determine how many of the following components are required to provide **whole-home backup for three days**:

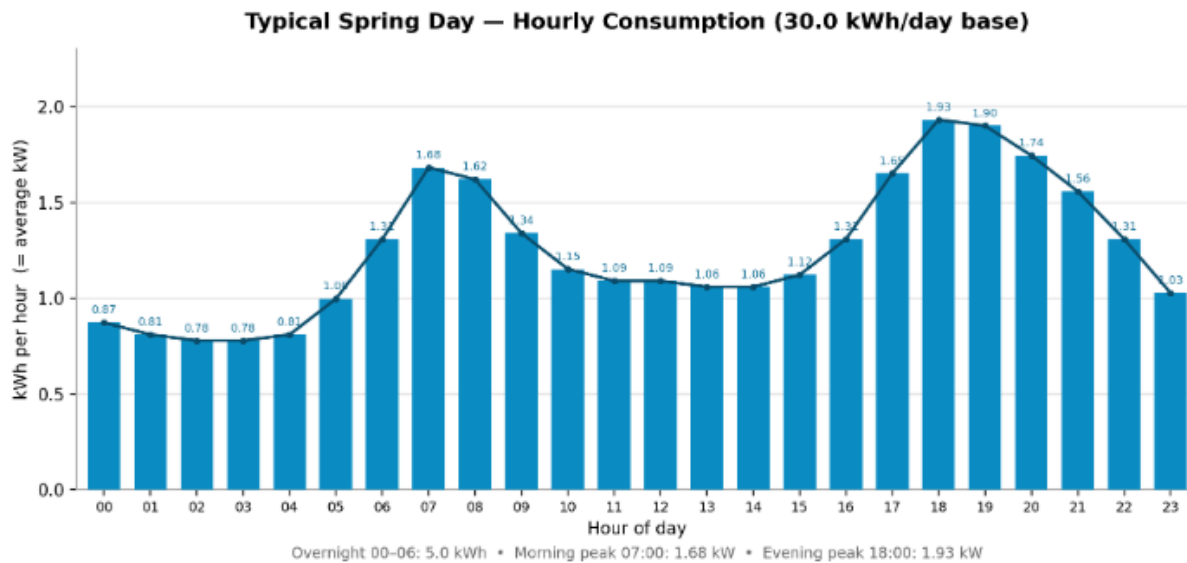
- Solar panels
- FlexBOSS21 inverters
- GridBOSS units
- EG4 280Ah batteries

Assume the house has a **springtime base consumption of 30 kWh per day**.

Seasonal energy consumption changes as follows:

- **Winter:** 30% increase from the spring baseline
- **Spring:** 30 kWh/day baseline
- **Summer:** 15% reduction from the spring baseline
- **Fall:** 15% increase from the spring baseline

Use the bell curve below to determine the approximate **hourly distribution of the home's electrical load throughout the day**.

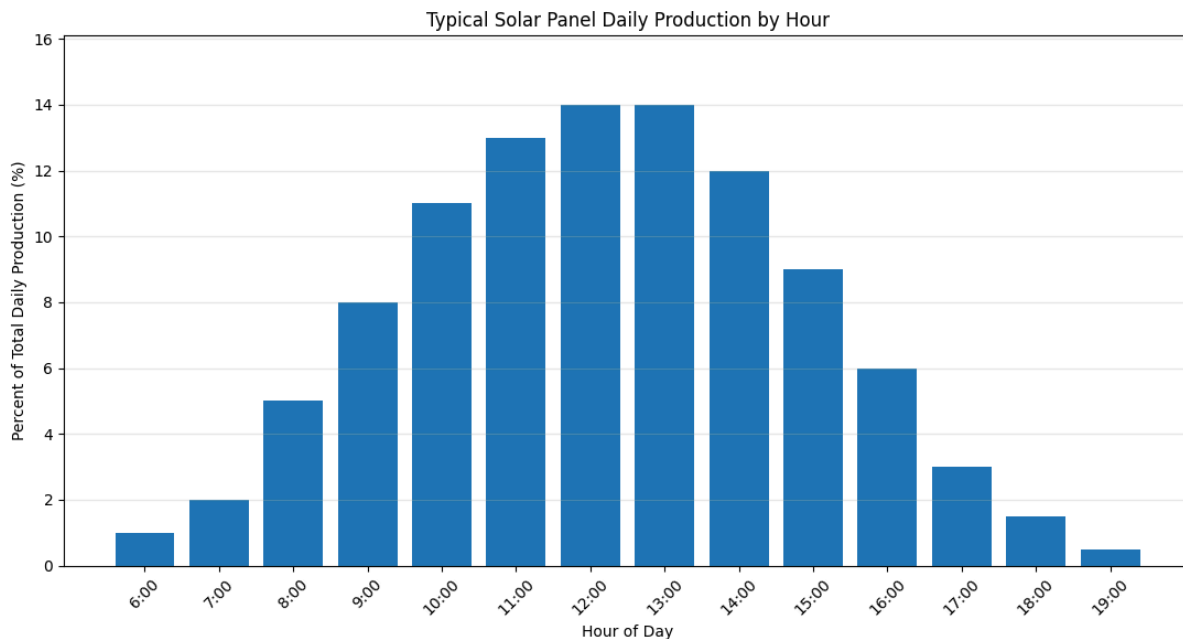


Solar Production Assumptions

For Victoria, BC, use the following average **equivalent peak sun hours per day**:

- **Winter:** 1.17 hours/day
- **Spring:** 4.25 hours/day
- **Summer:** 6.02 hours/day
- **Fall:** 2.41 hours/day

These values represent equivalent peak sun hours and should be used when estimating seasonal solar production.



Generator Optimization

Your group's goal is to take the system **completely off-grid while minimizing the annual operating cost of the generator.**

Assume the generator:

- Costs **\$1.65 per kWh** of electricity produced
- Requires servicing after every **100 kWh of generator production**
- Costs **\$300 per service**
- Experiences a **2% reduction in efficiency after each service**

Your design should determine how the generator can best support the solar and battery system while minimizing its annual use and associated servicing costs.



Consider how seasonal solar production, battery capacity, inverter capacity, and the home's hourly load profile affect when the generator may need to operate.

Helpful Calculations

Open-Circuit Voltage for MPPT Sizing

Use the following calculation when determining the maximum solar string voltage and verifying compatibility with the FlexBOSS21 MPPT operating limits:

1. Find the temperature difference

$$\Delta T = T_{min} - 25^{\circ}C$$

For temperatures below 25°C, this will be negative.

2. Calculate the Voc increase

Substitute $-0.250\%/^{\circ}C$ for the temperature coefficient:

$$\text{Voltage Increase (\%)} = \Delta T \times (-0.250)$$

or:

$$\text{Voltage Increase (\%)} = (T_{min} - 25) \times (-0.250)$$

Because both numbers are negative in cold weather, the resulting voltage change is positive.

3. Calculate maximum cold-weather Voc

$$V_{oc,max} = V_{oc} \times \left(1 + \frac{(T_{min} - 25) \times (-0.250)}{100} \right)$$



Equipment Costs

Use the following equipment costs when determining the total cost of your proposed system:

- **Distribution Splitter: \$4,500**
- **Fused Disconnect: \$1,500**
- **FlexBOSS21: \$5,500**
- **440W JA Solar Panel + Associated Racking: \$350 per panel**
- **280Ah EG4 Battery: \$5,500**
- **EG4 GridBOSS: \$2,500**
- **200A Electrical Panel: \$1,500**

Your final design should consider both system performance and overall project cost.

Equipment Explanations

Distribution Splitter

A distribution splitter is a connection point used to divide one incoming electrical supply into multiple outgoing feeders. Instead of routing all power through one panel first, the splitter allows several downstream pieces of equipment to be supplied from the same source.

Electrically, it acts as a common bus where conductors are joined while maintaining proper ampacity, insulation, enclosure, and fault-current ratings.



In larger systems, splitters are useful when one service needs to feed multiple panels, inverters, disconnects, or separate building loads.

Fused Disconnect

A fused disconnect combines two functions: it provides a visible means of electrically isolating equipment and uses fuses to protect downstream conductors and equipment from excessive current.

If a fault or overload causes current to exceed the fuse rating for long enough, the fuse opens the circuit. The disconnect portion allows technicians to safely shut off the circuit for maintenance.

Engineers need to consider fuse size, interrupting rating, conductor ampacity, available fault current, and coordination with other protective devices.

Main Distribution Panel

The main distribution panel receives electrical power and distributes it to multiple branch circuits or subpanels.

Internally, it normally contains bus bars, circuit breakers, neutral conductors, and grounding connections. The main breaker limits the maximum current supplied through the panel, while individual breakers protect each outgoing circuit.

From a design perspective, engineers need to check panel voltage, phase configuration, bus rating, breaker ratings, load calculations, and short-circuit ratings.

Transfer Switch



A transfer switch changes a load from one power source to another, commonly between the utility and a generator.

Its main purpose is to prevent two sources from being unintentionally connected together. For example, when utility power fails, a transfer switch can disconnect the utility source before connecting the generator.

Transfer switches can be manual or automatic. The switching sequence and interlocking are important because paralleling two unsynchronized AC sources can cause severe equipment damage and create significant safety hazards.

Emergency Loads Panel

An emergency loads panel supplies circuits that need to remain powered when the normal electrical supply is unavailable.

Typical examples may include emergency lighting, communications equipment, fire/life-safety systems, critical receptacles, pumps, or other selected building loads.

Instead of placing an entire building on backup power, engineers will often isolate the most important loads. This can significantly reduce the required generator, battery, inverter, and conductor sizes.

Generator

A generator converts mechanical energy into electrical energy.

In a typical standby system, an engine rotates an alternator, which creates AC voltage through electromagnetic induction.



The generator must be sized not only for steady-state loads but also for transient conditions such as motor starting currents.

Engineers also need to consider voltage regulation, frequency regulation, grounding, neutral configuration, fault current, fuel supply, and how the generator interfaces with the rest of the backup power system.

Inverter

An inverter converts DC electrical power into AC power.

In solar and battery systems, DC power from PV modules or batteries is electronically switched and controlled to produce an AC waveform compatible with the building's electrical system.

Modern inverters do much more than simple DC-to-AC conversion. They can regulate voltage and frequency, control real and reactive power, monitor the grid, limit export, charge batteries, and disconnect during abnormal grid conditions.

Inverter selection usually involves reviewing:

- DC voltage range
- MPPT limits
- AC voltage
- Output current
- Continuous power rating
- Surge power rating
- Efficiency
- Grid-interconnection requirements



Basic System Flow

A simple way to visualize a conventional electrical system is:

Utility → Fused Disconnect → Distribution Splitter → Main Distribution Panel(s)

A traditional backup system may look like:

Generator / Inverter → Transfer Switch → Emergency Loads Panel

The **transfer switch determines which power source supplies the emergency loads**, while the main distribution panel continues to serve the normal building loads during standard operating conditions.

For this challenge, your team will need to determine how the **solar array, batteries, FlexBOSS21 inverter(s), GridBOSS unit(s), generator, and existing 400A service** should be integrated to create a practical whole-home off-grid system.